

REQUIREMENT ANALYSIS DOCUMENT (RAD)

PROJECT TITLE:

Online Shopping Management System

1. Introduction

The Online Shopping Management System is a web-based application that enables customers to purchase products from anywhere using the internet. The system provides a user-friendly platform for customers to search products, compare prices, place orders, make secure payments, and track deliveries. It also allows administrators to manage customers, products, inventory, and orders efficiently.

The objective of the system is to provide a fast, reliable, and secure shopping experience while reducing manual work and improving business efficiency.

2. Objectives

The main objectives of the proposed system are:

- * Develop an online shopping platform for customers.
- * Maintain customer information securely.
- * Display products based on categories.
- * Process customer orders efficiently.
- * Support multiple online payment methods.
- * Track shipments from warehouse to customer.
- * Generate reports for business analysis.

3. Functional Requirements

3.1 Customer Registration

The system should allow new users to create an account by providing:

- * Full Name
- * Email Address

- * Mobile Number
- * Password
- * Delivery Address

Requirements

- * Email ID must be unique.
- * Password should contain at least 8 characters.
- * Customers can edit their profile anytime.

3.2 User Login

The system shall authenticate registered users using:

- * Email
- * Password

Requirements

- * Invalid login credentials should display an error.
- * Forgotten passwords can be reset using email verification.

3.3 Product Management

The system should display products with:

- * Product ID
- * Product Name
- * Category
- * Brand
- * Price
- * Description
- * Product Image
- * Stock Availability

Requirements:

- * Products should be grouped by category.

- * Customers can search using product name or category.
- * Customers can sort products by price or popularity.

3.4 Shopping Cart

Customers should be able to:

- * Add products
- * Remove products
- * Change quantity
- * View total amount

Requirements

- * Cart total should automatically update.
- * Customers can continue shopping after adding items.

3.5 Order Processing

Customers can place an order after confirming the cart.

Order information includes:

- * Order ID
- * Customer ID
- * Order Date
- * Product List
- * Quantity
- * Total Price
- * Order Status

Order Status:

- * Pending
- * Confirmed
- * Packed
- * Shipped

- * Delivered
- * Cancelled

3.6 Payment Management

Supported payment methods:

- * Credit Card
- * Debit Card
- * UPI
- * Net Banking
- * Cash on Delivery

Requirements:

- * Payment should be verified before confirming the order.
- * Failed transactions should not generate orders.

3.7 Shipment Tracking:

After successful payment, customers can track their order.

Shipment details include:

- * Tracking Number
- * Courier Partner
- * Dispatch Date
- * Expected Delivery Date
- * Current Location
- * Delivery Status

4. Non-Functional Requirements

Performance:

- * Product pages should load quickly.

- * Search results should appear within a few seconds.

Security:

- * Passwords should be encrypted.
- * Payment information should be securely stored.
- * Only authorized users can access administrative functions.

Reliability:

- * Database should maintain data integrity.
- * Daily backups should be taken.

Availability:

- * System should be available 24×7.

5. Users of the System:

Customer:

Functions:

- * Register
- * Login
- * Search products
- * Add to cart
- * Place orders
- * Make payments
- * Track shipments
- * View order history

Administrator

Functions:

- * Add products
- * Update products
- * Delete products
- * Manage customers
- * Process orders
- * Generate reports
- * Manage inventory

Delivery Partner

Functions:

- * View assigned deliveries
- * Update shipment status
- * Confirm successful delivery

6. Business Rules

- * Every customer has a unique Customer ID.
- * Every product has a unique Product ID.
- * Product price must always be greater than zero.
- * Stock quantity cannot be negative.
- * One customer can place multiple orders.
- * One order can contain multiple products.
- * Payment must be completed before shipment.
- * Cancelled orders cannot be shipped.

7. Reports Required

Customer Report

Displays:

- * Customer Name
- * Email
- * Total Orders
- * Total Amount Spent

Product Sales Report

Displays:

- * Product Name
- * Quantity Sold
- * Revenue Generated

Inventory Report

Displays:

- * Product Name
- * Available Stock
- * Low Stock Products

Order Status Report

Displays:

- * Order ID
- * Customer Name
- * Order Date
- * Payment Status
- * Delivery Status

8. Assumptions

- * Customers have internet access.
- * Online payment gateway is available.
- * Courier service is integrated with the system.
- * Products are maintained by administrators.

9. Conclusion

The Online Shopping Management System simplifies the complete shopping process by integrating customer management, product catalog, shopping cart, online payment, order processing, inventory control, and shipment tracking into a single

platform. The proposed system provides a secure, reliable, and efficient solution that improves customer satisfaction and business operations while supporting future scalability.